数据结构 A 第 3 次作业

姓名: 王子川　　**学号:** PB25000055

1(2.33) 链表的分割

程序用指针 p 遍历给定链表 L. 设当前结点的数据为字符 x, 根据 x 的类别, 将结点 p 接到相应链表 A, B, C 的尾部, 并更新相应的尾指针 a, b, c. 最后令三个尾结点的 next 域分别指向头结点 A, B, C, 从而形成三个循环链表.

```
Status LinkListDivision(LinkList L, LinkList& A, LinkList& B, LinkList& C)
    {
    if (L == NULL) return ERROR;
    // Create head nodes of A, B and C.
    A = new LNode;
    B = new LNode;
    C = new LNode;
    // a, b, c point to the tails of linked lists A, B, C.
    LinkList a = A, b = B, c = C;
    LinkList p = L->next;

    while (p) {
        // Save the next node in the original list.
        LinkList q = p->next;
        char x = p->data;

        if ((x >= 'a' && x <= 'z') || (x >= 'A' && x <= 'Z')) {
            a->next = p;
            a = p;
        }
        else if (x >= '0' && x <= '9') {
            b->next = p;
            b = p;
        }
        else {
            c->next = p;
            c = p;
        }
        p = q;
    }
    // Make A, B and C circular linked lists.
    a->next = A;
    b->next = B;
    c->next = C;
    L->next = NULL;
    return OK;
}
```

2(2.38) 命中访存

先遍历指针 p 找到 x, 接着把这个节点从双向循环链表 L 中取出, $freq$ 递增, 插入对应的位置.

```
Status Locate(DuLinkList& L, ElemType x) {
    if (L == NULL) return ERROR;
    DuLinkList p = L->next;

    // Find x.
    while ((p != L) && (p->data != x)) p = p->next;
    if (p == L) return ERROR;
    (p->freq)++;

    // Remove p.
    p->pre->next = p->next;
    p->next->pre = p->pre;
    DuLinkList q = p->pre;

    // find the new position.
    while ((q != L) && q->freq < p->freq) q = q->pre;

    // Insert p.
    p->pre = q;
    p->next = q->next;
    p->next->pre = p;
    q->next = p;

    return OK;
}
```